Exact one-body Green's functions and self-energies from exact many-body states

Intuition, method, implementation and testing of the exact G and Σ_{xc} solver in iDEA

Technical report accompanying the iDEA pull request

`iDEA.methods.interacting.green_function`, `iDEA.observables`, `iDEA.plotting`

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Abstract

This report describes how iDEA builds the *exact* one-body Green's function $G(x, x'; t \text{ or } \omega)$ and the *exact* exchange-correlation self-energy $\Sigma_{xc}(x, x'; t \text{ or } \omega)$ of an interacting electronic state. Both are built from the exact eigenstates of the $(N-1)$-, N- and $(N+1)$-electron systems.

- **The $N \pm 1$ problems.** These are posed in an ordered Slater-determinant basis. They are solved either by complete diagonalisation or by a block Krylov method seeded with the states $\hat{\psi}^{(\dagger)}(x)\,|\Psi_0\rangle$. The block Krylov method reproduces the low-order moments of G exactly. As a result, the sum rules and the static self-energy are exact even when the excitation spectrum is truncated.
- **The self-energy.** It is obtained in closed pole form by Löwdin partitioning, carried out as a block Lanczos recursion on the poles of G. Each correlation pole carries a *character* that fixes its time ordering.
- **Evaluation.** The resulting objects can be evaluated in real time, at real or complex frequency, and on the imaginary axes used by the GW code in iDEA, so exact and GW results can be compared directly. They can also be animated as videos of $G(x, x')$ and $\Sigma(x, x')$.

We explain the physical meaning of each element of G and Σ, derive every step of the method, describe the code module by module, and list the tests. On every grid tested, the implementation satisfies all exact identities (completeness, density matrix, first moment, Galitskii–Migdal energy and Dyson's equation) to machine precision.

As an illustration, we compute the exact G and Σ_{xc} of the one-dimensional hydrogen atom. The exact self-energy annihilates the occupied orbital at every frequency. Self-consistent GW instead leaves a self-screening residual of order 10^{-2}.

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1 Introduction

The one-body Green's function G is the central object of many-body perturbation theory [1–4]. It contains:

- the ground-state density and density matrix;
- the total energy, through the Galitskii–Migdal formula [5];
- the complete spectrum of electron addition and removal energies, i.e. photoemission and inverse photoemission.

The self-energy Σ is defined by Dyson's equation as the difference between the inverse of G and that of a simple reference. It is the non-local, energy-dependent potential that an added or removed electron feels. Approximations such as GW [1] are approximations to Σ. Judging and improving them requires knowing what the exact Σ looks like.

In real materials, the exact Σ is out of reach. In one-dimensional model systems, the exact many-body states can be found numerically for a few electrons on a fine real-space grid. This is what iDEA was built for [6–9].

This report describes a new part of iDEA that builds the *exact* G and *exact* Σ_{xc} from those states, with the full dependence on both positions x and x' and on time or frequency. To our knowledge, such exact two-point quantities have rarely been shown for real-space model systems. Being able to compute and *watch* them, as videos of $G(x, x'; t)$ or $\Sigma(x, x'; \omega)$, has three uses:

- a benchmark for GW and vertex corrections [8, 10, 11];
- a direct test of the nearsightedness of the self-energy [9, 12, 13], which is the subject of the accompanying research proposal;

- a teaching tool.

The report is organised as follows:

- Section 2 explains what each element of G and Σ means physically.
- Section 3 derives the method.
- Section 4 describes the code.
- Section 5 describes the tests.
- Section 6 shows illustrative results for the one-dimensional hydrogen atom.
- Section 7 lists limitations and next steps.
- Appendix A gives the production calculations to run on a GPU.
- Appendix B collects derivations.

Hartree atomic units are used throughout.

2 Background and intuition

2.1 The Green's function: propagation of an added electron or hole

For a state $|\Psi_0\rangle$ of energy E_0, the time-ordered Green's function of spin channel σ is

$$G_\sigma(x, x'; t) = -i \, \langle \Psi_0 | \mathcal{T} \, \hat{\psi}_\sigma(x, t) \, \hat{\psi}^\dagger_\sigma(x', 0) | \Psi_0 \rangle \, , \tag{1}$$

where $\hat{\psi}_\sigma(x, t) = e^{i\hat{H}t} \hat{\psi}_\sigma(x) e^{-i\hat{H}t}$.

Positive times: an added electron. For $t > 0$,

$$iG(x, x'; t) = \langle \Psi_0 | \hat{\psi}(x) \, e^{-i(\hat{H} - E_0)t} \, \hat{\psi}^\dagger(x') | \Psi_0 \rangle \, . \tag{2}$$

In words:

1. add an electron at x';
2. let the whole interacting $(N + 1)$-electron system evolve for a time t;
3. project onto the state in which an electron has been added at x to the undisturbed ground state.

A column $G(\cdot, x_0; t)$ is therefore the wavefunction of an electron injected at x_0, dressed by the response of all the others.

Negative times: a hole. For $t < 0$, the same holds for a hole: an electron removed, as in photoemission.

Equal times: the density matrix. The equal-time limit from below is the one-body density matrix,

$$G_\sigma(x, x'; 0^-) = i\gamma_\sigma(x, x'), \qquad \gamma_\sigma(x, x') = \langle \Psi_0 | \hat{\psi}^\dagger_\sigma(x') \hat{\psi}_\sigma(x) | \Psi_0 \rangle \, . \tag{3}$$

- Its diagonal is the spin density, $n_\sigma(x) = \gamma_\sigma(x, x)$. This is the precise sense in which the density is the diagonal of G.
- Off the diagonal, $\gamma(x, x')$ is the coherence between removing an electron at x and removing it at x'. It determines the kinetic energy, the exchange energy and the momentum distribution.
- How fast γ decays with $|x - x'|$ is one measure of nearsightedness [12, 13].

2.2 Lehmann representation, Dyson orbitals and the spectral function

Inserting complete sets of $(N \pm 1)$-electron eigenstates into Equation (1) gives the Lehmann representation [2, 14]:

$$G_\sigma(x, x'; z) = \sum_k \frac{d_k(x)\, d_k^*(x')}{z - \varepsilon_k}. \tag{4}$$

There are two kinds of pole:

- **Removal poles,** $\varepsilon_k = E_0 - E_k^{N-1}$, with Dyson orbitals $d_k(x) = \langle \Psi_k^{N-1} | \hat{\psi}_\sigma(x) | \Psi_0 \rangle$.
- **Addition poles,** $\varepsilon_k = E_k^{N+1} - E_0$, with Dyson orbitals $d_k(x) = \langle \Psi_0 | \hat{\psi}_\sigma(x) | \Psi_k^{N+1} \rangle$.

The poles are the energies measured in photoemission ($-\varepsilon_k$ is an ionisation energy) and inverse photoemission. Each Dyson orbital is the wavefunction of the electron removed or added, and its norm is the spectroscopic factor or quasiparticle weight.

Time and frequency forms. The time-ordered function is

$$G(x, x'; t) = -i \sum_k d_k(x) d_k^*(x')\, e^{-i\varepsilon_k t} \big[\theta(t)(1 - f_k) - \theta(-t) f_k\big], \tag{5}$$

where $f_k = 1$ for removal and $f_k = 0$ for addition poles. In frequency, it is Equation (4) with $z = \omega + i\eta$ for addition poles and $z = \omega - i\eta$ for removal poles, $\eta \to 0^+$.

The spectral function. It is

$$A(x, x'; \omega) = \sum_k d_k(x) d_k^*(x')\, \delta(\omega - \varepsilon_k). \tag{6}$$

Its diagonal, $A(x, x; \omega)$, is the local density of states for adding or removing an electron of energy ω at x, which is what a scanning-tunnelling microscope measures.

Sum rules. The anticommutation relations and the Hamiltonian give exact sum rules:

$$\int A(x, x'; \omega)\, d\omega = \delta(x - x') \qquad \text{(completeness)}, \tag{7}$$

$$\int_{-\infty}^{\mu} A(x, x'; \omega)\, d\omega = \gamma(x, x') \qquad \text{(density matrix)}, \tag{8}$$

$$\int \omega\, A(x, x'; \omega)\, d\omega = h_\infty(x, x') \qquad \text{(first moment)}, \tag{9}$$

$$E_0 = \tfrac{1}{2} \sum_\sigma \sum_{m \in \text{removal}} \langle d_m | \hat{h}_0 + \varepsilon_m | d_m \rangle \qquad \text{(Galitskii–Migdal)}. \tag{10}$$

The first-moment matrix is the static Hamiltonian

$$h_\infty(x, x') = h_0(x, x') + v_H(x)\delta(x - x') + \Sigma_{x\sigma}(x, x'), \qquad \Sigma_{x\sigma}(x, x') = -\gamma_\sigma(x, x')\, v(x, x'), \tag{11}$$

where $h_0 = -\frac{1}{2}\partial_x^2 + v_{\text{ext}}$, v_H is the Hartree potential of the total density, and Σ_x is the Fock operator of the *exact* density matrix (Appendix B.3).

2.3 The self-energy: the potential felt by the added or removed electron

The exchange-correlation self-energy is defined by Dyson's equation,

$$G^{-1}(z) = z - h_0 - v_H - \Sigma_{xc}(z), \qquad \Sigma_{xc}(z) = \Sigma_x + \Sigma_c(z). \tag{12}$$

The quasiparticle equation. Because $G^{-1}(\varepsilon_k)\, d_k = 0$, the Dyson orbitals satisfy the quasiparticle equation exactly:

$$[h_0 + v_H]\, d_k(x) + \int \Sigma_{xc}(x, x'; \varepsilon_k)\, d_k(x')\, dx' = \varepsilon_k\, d_k(x). \tag{13}$$

So Σ_{xc} plays the role that the Kohn–Sham v_{xc} plays for Kohn–Sham orbitals [15], with two differences:

- it is *non-local*: the element (x, x') moves amplitude from x' to x;
- it depends on *energy*: an electron added at a different energy feels a different potential.

The static part. Expanding Equation (4) at large z gives $G(z) = z^{-1} + h_\infty z^{-2} + O(z^{-3})$. Hence $G^{-1}(z) = z - h_\infty + O(z^{-1})$, and

$$\Sigma_{xc}(z \to \infty) = \Sigma_x. \tag{14}$$

The static part of the exact self-energy is the Fock operator built from the exact density matrix. It is instantaneous in time, carries Pauli exclusion and the self-interaction correction, and has the spatial range of γ.

The correlation part. The correlation part has poles,

$$\Sigma_c(x, x'; z) = \sum_p \frac{r_p(x)\, r_p^*(x')}{z - \zeta_p}. \tag{15}$$

Each part has a physical reading:

- **Poles ζ_p.** These are the energies at which the added or removed electron can hand energy to the rest of the system by exciting it (two-particle–one-hole and two-hole–one-particle configurations, or shake-up). They produce satellites in $A(\omega)$ and take weight away from the quasiparticle peaks.
- **Residues $r_p(x)$.** These say how strongly an electron at x couples into each such channel.
- $\mathrm{Re}\,\Sigma_c$**.** This is the relaxation and polarisation energy.
- $-\mathrm{Im}\,\Sigma_c^R(x, x; \omega) \geq 0$**.** This acts like a local absorbing potential: the rate at which an electron added with energy ω leaks out of the one-particle channel.
- $\Sigma_c(x, x'; t)$ **in real time.** This is a memory kernel: the amplitude for the electron to leave the one-particle channel at x' (by exciting the others) and to return at x a time t later. In GW, $\Sigma = iGW$: the electron interacts with the polarisation it has left behind.

2.4 One electron: no self-interaction, and *GW* self-screening

For a single spin-up electron in the orbital ϕ, with $\hat{h}_0\phi = E_0\phi$, the structure is very simple.

The removal part. The only $(N-1)$-electron state is the vacuum. So $G_\uparrow$ has a single removal pole, at $\varepsilon = E_0$, with Dyson orbital ϕ and weight one. By completeness (Equation (7)), all addition Dyson orbitals lie in the orthogonal complement of ϕ. Therefore $G_\uparrow(z) = \phi\phi^*/(z - E_0) \oplus G_{\mathrm{add}}(z)$, and $G_\uparrow^{-1}(z)\phi = (z - E_0)\phi$ at every z.

The self-energy. Inserting this into Equation (12) with $\hat{h}_0\phi = E_0\phi$ gives

$$[v_H + \Sigma_{xc\uparrow}(z)]\phi = 0 \quad \text{for all } z. \tag{16}$$

The exchange part does this alone, since $\Sigma_{x\uparrow}\phi = -v_H\phi$ when $\gamma_\uparrow = \phi\phi^*$. So the correlation part must annihilate the orbital: $\Sigma_{c\uparrow}(z)\phi = 0$. The exact self-energy has no self-interaction at any frequency.

The *GW* self-energy. The GW self-energy iGW contains the electron's own orbital in G, and W screens the electron by its own density. This leaves $\Sigma_c^{GW}\phi \neq 0$: the electron screens itself. This self-screening error was identified in the hydrogen atom by Nelson *et al.* [11] and analysed with iDEA in Ref. [8].

Spin down. In the spin-down channel, $G_\downarrow$ has no removal part at all, $\gamma_\downarrow = 0$ and $\Sigma_{x\downarrow} = 0$. So $\Sigma_\downarrow = \Sigma_{c\downarrow}$ is pure correlation: it describes how an added spin-down electron polarises the bound spin-up electron. This is the physics that binds the H^- ion.

3 Method

3.1 Discretisation and second quantisation on the grid

The grid and field operators. iDEA represents states on a uniform grid x_i, $i = 1, \ldots, N_x$, with spacing δx. We define dimensionless fermion operators

$$c_{i\sigma} = \sqrt{\delta x}\,\hat{\psi}_\sigma(x_i), \qquad \{c_{i\sigma}, c^\dagger_{j\sigma'}\} = \delta_{ij}\delta_{\sigma\sigma'}. \tag{17}$$

Kernels and matrices. Two-point quantities are stored as *kernels* $A(x_i, x_j)$, which act as $\int A(x,x')f(x')\,dx' \to \sum_j A(x_i,x_j)f(x_j)\,\delta x$. The matrix that acts on discrete vectors is therefore $\mathsf{A} = A\,\delta x$. This is the convention of the GW module `iDEA.methods.mbpt` [16, 17], and it is used throughout.

The Hamiltonian. It is

$$\hat{H} = \sum_{ij\sigma} (\mathsf{h}_0)_{ij}\, c^\dagger_{i\sigma} c_{j\sigma} + \frac{1}{2}\sum_{ij\sigma\sigma'} v_{ij}\, c^\dagger_{i\sigma} c^\dagger_{j\sigma'} c_{j\sigma'} c_{i\sigma}, \tag{18}$$

- h_0 is the finite-difference kinetic-energy stencil plus $\mathrm{diag}(v_{\mathrm{ext}})$;
- $v_{ij} = v(x_i, x_j)$ is the (softened) interaction.

The many-body state. An iDEA many-body state stores the full antisymmetric wavefunction $\Psi(x_1\sigma_1, \ldots, x_N\sigma_N)$, normalised as $\sum |\Psi|^2 \delta x^N = 1$. We work with the *spin-ordered* spatial wavefunction, with all $N_\uparrow$ spin-up electrons first:

$$\psi(x_1, \ldots, x_{N_\uparrow}; y_1, \ldots, y_{N_\downarrow}) = \sqrt{\tbinom{N}{N_\uparrow}}\;\delta x^{N/2}\;\Psi(x_1\uparrow, \ldots, x_{N_\uparrow}\uparrow, y_1\downarrow, \ldots, y_{N_\downarrow}\downarrow). \tag{19}$$

It is antisymmetric within each spin group and normalised as $\sum |\psi|^2 = 1$ (Appendix B.1). The code checks this normalisation and raises an error if it fails.

3.2 Sector Hamiltonians in an ordered Slater-determinant basis

The $(N \pm 1)$-electron problems are solved in *sectors* with fixed numbers $(n_\uparrow, n_\downarrow)$ of up and down electrons.

The determinant basis. A sector is spanned by the ordered determinants

$$|\mathbf{a};\mathbf{b}\rangle = c^\dagger_{a_1\uparrow}\cdots c^\dagger_{a_{n_\uparrow}\uparrow} c^\dagger_{b_1\downarrow}\cdots c^\dagger_{b_{n_\downarrow}\downarrow}|0\rangle\,, \qquad a_1 < \cdots < a_{n_\uparrow}, \quad b_1 < \cdots < b_{n_\downarrow}. \tag{20}$$

Sites are labelled $0, \ldots, N_x - 1$. The combinations are enumerated in colexicographic order, so the position of a combination in the list equals its rank in the combinatorial number system:

$$r(\mathbf{a}) = \sum_{j=1}^{n} \binom{a_j}{j}. \tag{21}$$

Table 1: Dimensions of the $(N \pm 1)$-electron sectors. Hydrogen ($N = 1$, spin up) needs the vacuum and the sectors $(2, 0)$ and $(1, 1)$. A two-electron atom ($N_\uparrow = N_\downarrow = 1$) needs $(0, 1)$, $(1, 0)$, $(2, 1)$ and $(1, 2)$.

system	N_x	sector	D	product space $N_x^{N\pm1}$
hydrogen	61	$(2, 0)$	1 830	3 721
hydrogen	61	$(1, 1)$	3 721	3 721
hydrogen	151	$(2, 0)$	11 325	22 801
hydrogen	151	$(1, 1)$	22 801	22 801
two electrons	21	$(2, 1)$	4 410	9 261
two electrons	41	$(2, 1)$	33 620	68 921
two electrons	101	$(2, 1)$	510 050	1 030 301

The determinant $|\mathbf{a}; \mathbf{b}\rangle$ has index $r(\mathbf{a})\binom{N_x}{n_\downarrow} + r(\mathbf{b})$. Looking up the index of any determinant is therefore a vectorised sum over a precomputed table of binomial coefficients; no dictionaries or searches are needed.

Coefficients of a state. For a state with spin-ordered wavefunction ψ,

$$\langle \mathbf{a}; \mathbf{b}|\Psi\rangle = \sqrt{n_\uparrow!\, n_\downarrow!}\, \psi(\mathbf{a}; \mathbf{b}). \tag{22}$$

The isometry P. The existing iDEA Hamiltonian acts on the spatial product space $\mathbb{R}^{N_x^N}$ (C order, axis k for electron k). To reuse it unchanged, we build the sparse isometry P from the determinant basis into the product space. Column $(\mathbf{a}, \mathbf{b})$ of P is

$$P\, e_{(\mathbf{a},\mathbf{b})} = \frac{1}{\sqrt{n_\uparrow!\, n_\downarrow!}} \sum_{\pi \in S_{n_\uparrow}} \sum_{\rho \in S_{n_\downarrow}} \operatorname{sgn}(\pi) \operatorname{sgn}(\rho)\; e_{(a_{\pi(1)},\ldots,a_{\pi(n_\uparrow)},\, b_{\rho(1)},\ldots,b_{\rho(n_\downarrow)})}, \tag{23}$$

which satisfies $P^\top P = 1$. The sector Hamiltonian is then

$$\mathsf{H}_D = P^\top \mathsf{H}_{\text{prod}}\, P, \tag{24}$$

where H_{prod} is `interacting.hamiltonian` for a system with the same grid, potentials and stencil, and electrons `'u'*n_up + 'd'*n_dn`.

Dimensions. The sector has dimension $D = \binom{N_x}{n_\uparrow}\binom{N_x}{n_\downarrow}$, about $n_\uparrow!\, n_\downarrow!$ times smaller than N_x^N. Table 1 lists the sizes that matter here. The vacuum sector is handled separately: its energy is 0 and its only state is $|0\rangle$.

3.3 Field operators and Jordan–Wigner signs

Action on determinants. With the operator ordering above (all up operators to the left of all down operators), the annihilation operators act on a determinant as follows [18]:

$$c_{i\uparrow}\,|\mathbf{a}; \mathbf{b}\rangle = (-1)^{k-1}\,|\mathbf{a} \setminus a_k; \mathbf{b}\rangle \quad (i = a_k), \qquad c_{i\downarrow}\,|\mathbf{a}; \mathbf{b}\rangle = (-1)^{n_\uparrow+k-1}\,|\mathbf{a}; \mathbf{b} \setminus b_k\rangle \quad (i = b_k), \tag{25}$$

and give zero if site i is empty. The creation operators are the adjoints.

Source vectors. We need the N_x removal sources $g_i = c_{i\sigma}\,|\Psi\rangle$, which lie in the $(N-1)$ sector, and the N_x addition sources $f_i = c_{i\sigma}^\dagger\,|\Psi\rangle$, which lie in the $(N+1)$ sector. All of them are built at once:

1. Loop over the slot k in the larger of the two sectors.

2. For every determinant of that sector at once, delete slot k and find the rank of the smaller determinant from Equation (21).
3. Scatter the signed coefficient into the column for site $i = a_k$ (or b_k).

The product space is never used.

Checks. Two identities check the signs and the normalisation:

$$\langle g_i | g_j \rangle = \gamma(x_j, x_i)\,\delta x, \qquad \langle f_i | f_j \rangle = \delta_{ij} - \gamma(x_i, x_j)\,\delta x. \tag{26}$$

3.4 Lehmann poles

Each sector is diagonalised as $\mathsf{H}_D = \sum_n E_n \,|n\rangle\,\langle n|$.

Discrete Dyson amplitudes. These are:

- removal: $\varepsilon_n = E_0 - E_n$ and $\mathsf{V}_{in} = \langle n | g_i \rangle$;
- addition: $\varepsilon_n = E_n - E_0$ and $\mathsf{V}_{in} = \langle n | f_i \rangle^*$.

With these,

$$\mathsf{G}_{ij}(z) = \sum_n \frac{\mathsf{V}_{in}\mathsf{V}_{jn}^*}{z - \varepsilon_n}, \qquad d_n(x_i) = \mathsf{V}_{in}/\sqrt{\delta x}. \tag{27}$$

Practical details.

- **Unreachable states.** States with total weight $\sum_i |\mathsf{V}_{in}|^2 < 10^{-14}$ cannot be reached by the field operator (for example, because of symmetry) and are dropped.
- **The ground-state energy.** E_0 is recomputed as a Rayleigh quotient in the determinant basis, and its residual $\|(\mathsf{H}_D - E_0)\Psi\|$ is checked: the code raises an error above 10^{-6} and warns above 10^{-8}. This stops the tolerance of the N-electron eigensolver from shifting every pole.
- **The chemical potential.** A single $\mu = \frac{1}{2}(\min \varepsilon^+ + \max \varepsilon^-)$, at the centre of the gap over both spin channels, is used for the imaginary axis. It is `None` when there is no gap; for example, the lowest triplet state of a two-electron atom is an excited state with $\varepsilon^+ < \varepsilon^-$.
- **Derived quantities.** The density matrix $\gamma_\sigma = \sum_{\text{removal}} d\,d^*$, the density, the Hartree potential and the Galitskii–Migdal energy (Equation (10)) follow directly from the poles.

3.5 Large sectors: block Krylov with exact moments

Complete diagonalisation costs $O(D^3)$ time and $O(D^2)$ memory. Sectors larger than `dense_max` (default 6000) are therefore treated with a block Krylov (block Rayleigh–Ritz) method seeded with the sources $\mathsf{B} = [g_1, \ldots, g_{N_x}]$ or $[f_1, \ldots, f_{N_x}]$ [19, 20].

Building the basis. An orthonormal basis Q of

$$\mathcal{K}_m(\mathsf{H}_D, \mathsf{B}) = \operatorname{span}\{\mathsf{B}, \mathsf{H}_D\mathsf{B}, \ldots, \mathsf{H}_D^{m-1}\mathsf{B}\} \tag{28}$$

is built block by block:

- The first block is the left singular vectors of B whose singular values exceed $\tau\,\sigma_{\max}$.
- Each new block $\mathsf{W} = \mathsf{H}_D\mathsf{Q}_{\text{last}}$ is orthogonalised against all previous vectors by classical Gram–Schmidt applied twice (CGS2, which keeps orthogonality at the level of the unit round-off [21]).

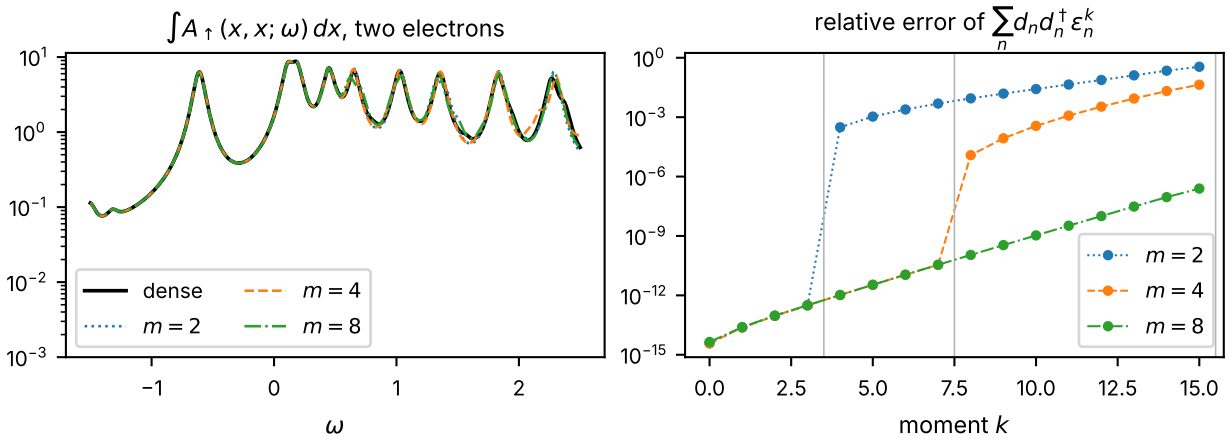


Figure 1: Block Krylov against complete diagonalisation for a two-electron atom ($v_{\text{ext}} = -2/(|x|+1)$, $N_x = 21$; the $(N+1)$ sectors have dimension 4,410). Left: the spectral function converges as the number of blocks m grows. Right: the moments $\sum_n d_n d_n^\dagger \varepsilon_n^k$ are exact to machine precision for every $k \le 2m-1$ (grey lines), as the theorem of Section 3.5 guarantees.

- The new block is then deflated by a thin SVD, keeping directions with singular values above $\tau\|\mathsf{H}_D\|_\infty$ (default $\tau = 10^{-10}$).

The recursion stops when `krylov_dim` vectors are reached (default $30N_x$) or when the space is exhausted.

The approximate poles. The projected matrix $\mathsf{T} = \mathsf{Q}^\dagger \mathsf{H}_D \mathsf{Q}$ (symmetrised) is diagonalised, $\mathsf{T} = \mathsf{Y}\Theta\mathsf{Y}^\dagger$. The Ritz values Θ give the poles and $\mathsf{Y}^\dagger\mathsf{Q}^\dagger\mathsf{B}$ the amplitudes.

Theorem (moment preservation). *For every $k \le 2m-1$, $\mathsf{B}^\dagger \mathsf{H}_D^k \mathsf{B} = \mathsf{B}^\dagger \mathsf{Q}\, \mathsf{T}^k\, \mathsf{Q}^\dagger \mathsf{B}$.*

Proof. Let $\Pi = \mathsf{Q}\mathsf{Q}^\dagger$. For $j \le m-1$, $\mathsf{H}_D^j \mathsf{B} \in \mathcal{K}_m$, and by induction $\mathsf{Q}\mathsf{T}^j\mathsf{Q}^\dagger\mathsf{B} = \Pi\mathsf{H}_D\Pi\mathsf{H}_D^{j-1}\mathsf{B} = \Pi\mathsf{H}_D^j\mathsf{B} = \mathsf{H}_D^j\mathsf{B}$. Write $k = a + c + b$ with $a, b \le m-1$ and $c \in \{0,1\}$. Then

$$\mathsf{B}^\dagger\mathsf{H}_D^k\mathsf{B} = (\mathsf{H}_D^a\mathsf{B})^\dagger\mathsf{H}_D^c(\mathsf{H}_D^b\mathsf{B}) = \mathsf{B}^\dagger\mathsf{Q}\mathsf{T}^a(\mathsf{Q}^\dagger\mathsf{H}_D^c\mathsf{Q})\mathsf{T}^b\mathsf{Q}^\dagger\mathsf{B} = \mathsf{B}^\dagger\mathsf{Q}\mathsf{T}^k\mathsf{Q}^\dagger\mathsf{B}. \qquad \square$$

Consequences. The moments of the Krylov approximation to G, $\sum_n \mathsf{V}_n\mathsf{V}_n^\dagger\varepsilon_n^k$, are polynomials of degree k in H_D, so they are exact for $k \le 2m-1$. This is the matrix form of Gauss quadrature [22]. Three things follow:

1. Completeness ($k = 0$) and the first moment ($k = 1$) are exact for *any* truncation.
2. So the static self-energy Σ_x (Equation (11)) is exact for any truncation too.
3. Truncation moves only the high-lying poles, and the pole spectrum converges as m grows. All poles are exact once the space is exhausted.

Figure 1 demonstrates this. The same moment-conserving idea underlies the block-Lanczos treatment of self-energies in auxiliary second-order Green's function methods [23].

GPU. With `GPU=True`, the dense diagonalisation (`cupy.linalg.eigh`) and the Krylov recursion (CuPy sparse matrix products, CGS2 and SVD) run on the GPU. The results are copied back to the host.

3.6 The exact self-energy in pole form

Collect the poles of one spin channel in $\mathsf{E} = \mathrm{diag}(\varepsilon_1, \ldots, \varepsilon_M)$ and the discrete amplitudes in $\mathsf{V} \in \mathbb{C}^{N_x \times M}$. Then $\mathsf{G}(z) = \mathsf{V}(z - \mathsf{E})^{-1}\mathsf{V}^\dagger$, and completeness says $\mathsf{V}\mathsf{V}^\dagger = 1$. Numerically this

holds to about 10^{-15}; a symmetric re-orthonormalisation $\mathsf{V} \leftarrow (\mathsf{V}\mathsf{V}^\dagger)^{-1/2}\mathsf{V}$ makes it exact without changing the span.

Löwdin partitioning. Complete the rows of V to a unitary matrix $\mathsf{U} = \binom{\mathsf{V}}{\mathsf{W}}$ and set $\mathsf{A} = \mathsf{U}\mathsf{E}\mathsf{U}^\dagger$. Then $\mathsf{G}(z) = [(z - \mathsf{A})^{-1}]_{11}$, and the Schur complement (Löwdin partitioning [24]) gives

$$\mathsf{G}^{-1}(z) = z - \mathsf{A}_{11} - \mathsf{A}_{12}(z - \mathsf{A}_{22})^{-1}\mathsf{A}_{21}. \tag{29}$$

Comparing with Equation (12):

$$\mathsf{A}_{11} = \mathsf{V}\mathsf{E}\mathsf{V}^\dagger = \mathsf{h}_0 + \mathrm{diag}(v_H) + \Sigma_x \delta x, \qquad \Sigma_c(z)\,\delta x = \mathsf{A}_{12}(z - \mathsf{A}_{22})^{-1}\mathsf{A}_{21}. \tag{30}$$

The self-energy does not depend on how the complement W is chosen.

Block Lanczos on the poles. We choose the completion by block Lanczos on the diagonal matrix E, starting from the orthonormal block $\mathsf{V}^\dagger$. This is the routine of Section 3.5 applied to E, which costs $O(M)$ per vector. It produces a block-tridiagonal $\mathsf{T} = \mathsf{Q}^\dagger\mathsf{E}\mathsf{Q}$ in which:

- the first diagonal block is $h_\infty \delta x$, so that $\Sigma_x = (h_\infty \delta x - \mathsf{h}_0 - \mathrm{diag}(v_H))/\delta x$;
- only the second block couples to the first.

Directions of $\mathbb{C}^M$ outside the Krylov space are invariant under E and orthogonal to $\mathsf{V}^\dagger$. They do not couple to the grid and are dropped exactly.

The poles of Σ_c. Diagonalising the remaining block, $\mathsf{T}_{22} = \mathsf{Z}\,\mathrm{diag}(\zeta)\,\mathsf{Z}^\dagger$, gives Equation (15) with

$$r_p = \mathsf{T}_{12}\mathsf{Z}_{:,p}/\sqrt{\delta x}. \tag{31}$$

A spin channel with M poles of G therefore has at most $M - N_x$ poles of Σ_c.

The check on the first block. The first block is checked against the Fock operator of the density matrix computed independently from the removal poles (Equation (11)). The largest deviation is stored in `info` as the first-moment error. A QR-based completion of V is kept in the tests as an independent reference for Equation (30).

3.7 Time ordering of the self-energy poles

The time-ordered G assigns $+i\eta$ to addition poles and $-i\eta$ to removal poles: $\mathsf{G}^T(\omega) = \mathsf{V}(\omega - \mathsf{E} + i\eta\mathsf{S})^{-1}\mathsf{V}^\dagger$, with $\mathsf{S} = \mathrm{diag}(\pm 1)$. Repeating the partition with the same U, to first order in η, replaces A_{22} by $\mathsf{A}_{22} - i\eta\,\mathsf{W}\mathsf{S}\mathsf{W}^\dagger$. First-order perturbation theory for the pole ζ_p then gives

$$\zeta_p \to \zeta_p - i\eta\, s_p, \qquad s_p = \sum_n S_n\,|\mathsf{Y}_{np}|^2 \in [-1, 1], \qquad \mathsf{Y} = \mathsf{Q}_{\geq 2}\mathsf{Z}. \tag{32}$$

- **The character.** s_p is the weight of the pole on addition poles of G minus its weight on removal poles. A pole is particle-like (it carries $+i\eta$ and $\theta(t)$) if $s_p > 0$, and hole-like if $s_p < 0$.
- **Degenerate poles.** Within degenerate clusters of ζ, the basis that diagonalises S is chosen first, as in degenerate perturbation theory.
- **Ambiguous poles.** If $|s_p| < 10^{-6}$, the time ordering is not determined at first order. This happens when a pole is built equally from addition and removal poles. The code warns and breaks the tie by the position of ζ_p relative to μ.

Table 2: Pole factors F_n on each axis, as implemented in `observables._pole_factors`. Here $f_n^+ = 1$ for particle-like poles (addition poles of G; $s_p > 0$ for Σ_c) and 0 otherwise.

axis	meaning	factor F_n
`t`	real time, time-ordered	$-ie^{-i\varepsilon_n t}\,[\theta(t)f_n^+ - \theta(-t)(1-f_n^+)]$
`t`	real time, retarded / advanced	$-ie^{-i\varepsilon_n t}\theta(t)$ / $ie^{-i\varepsilon_n t}\theta(-t)$
`omega`	real frequency, time-ordered	$f_n^+/(\omega - \varepsilon_n + i\eta) + (1-f_n^+)/(\omega - \varepsilon_n - i\eta)$
`omega`	real frequency, retarded / advanced	$1/(\omega - \varepsilon_n \pm i\eta)$
`z`	complex frequency	$1/(z - \varepsilon_n)$
`nu`	imaginary frequency about μ	$1/(\mu + i\nu - \varepsilon_n)$
`tau`	imaginary time	$-f_n^+ e^{-\lvert\varepsilon_n-\mu\rvert\tau}$ for $\tau > 0$; $(1-f_n^+)e^{-\lvert\varepsilon_n-\mu\rvert\lvert\tau\rvert}$ for $\tau \le 0$

Why not use ζ versus μ. The more familiar rule, retarded above μ and advanced below, holds for the continuous spectra of extended systems. It is *not* reliable for the discrete in-gap poles of finite systems: in a randomised test during development it gave the wrong ordering for 348 of 4780 in-gap poles. The character rule is checked against Dyson inversion of the time-ordered G in the tests (Section 5).

Imaginary time. On the imaginary-time axis, the pole position relative to μ is used instead, following the GW module. There, a decaying representation requires it.

3.8 Evaluation on time and frequency axes

The pole factors. Given poles ε_n with particle weights $f_n^+ \in \{0, 1\}$, the code evaluates $\sum_n d_n d_n^\dagger F_n$ with the factors F_n of Table 2. The same table serves G (using its occupations) and Σ_c (using the characters).

Conventions at equal time. The equal-time limit is taken as $t = 0^-$, so that $G(t{=}0) = i\gamma$ and $G(\tau{=}0) = \gamma$. These are the conventions of `iDEA.methods.mbpt`, so exact and GW quantities can be laid on the same grids.

Other quantities.

- **The spectral function** is evaluated directly as a sum of Lorentzians of width η.
- **Dyson inversion.** `self_energy(..., method="dyson")` inverts Equation (12) at each frequency, $\Sigma_{xc} = [w - \mathsf{h}_0 - \mathrm{diag}(v_H) - (\mathsf{G}\delta x)^{-1}]/\delta x$. This gives an independent check of the pole form.
- **The Galitskii–Migdal energy** uses Equation (10).

3.9 Visualisation

`iDEA.plotting` draws any two-point quantity $A(x, x')$ as side-by-side heatmaps of its real, imaginary, absolute and phase parts, and animates it over the axis.

Colour scales. The scale is fixed across all frames. By default it is the 99.5th percentile of the magnitude over the whole video, linear or symmetric-logarithmic, and symmetric about zero for real and imaginary parts. A strip under the heatmaps can show a one-dimensional trace with a moving marker:

- $\mathrm{Tr}\,A(\omega)$ for real-frequency G;

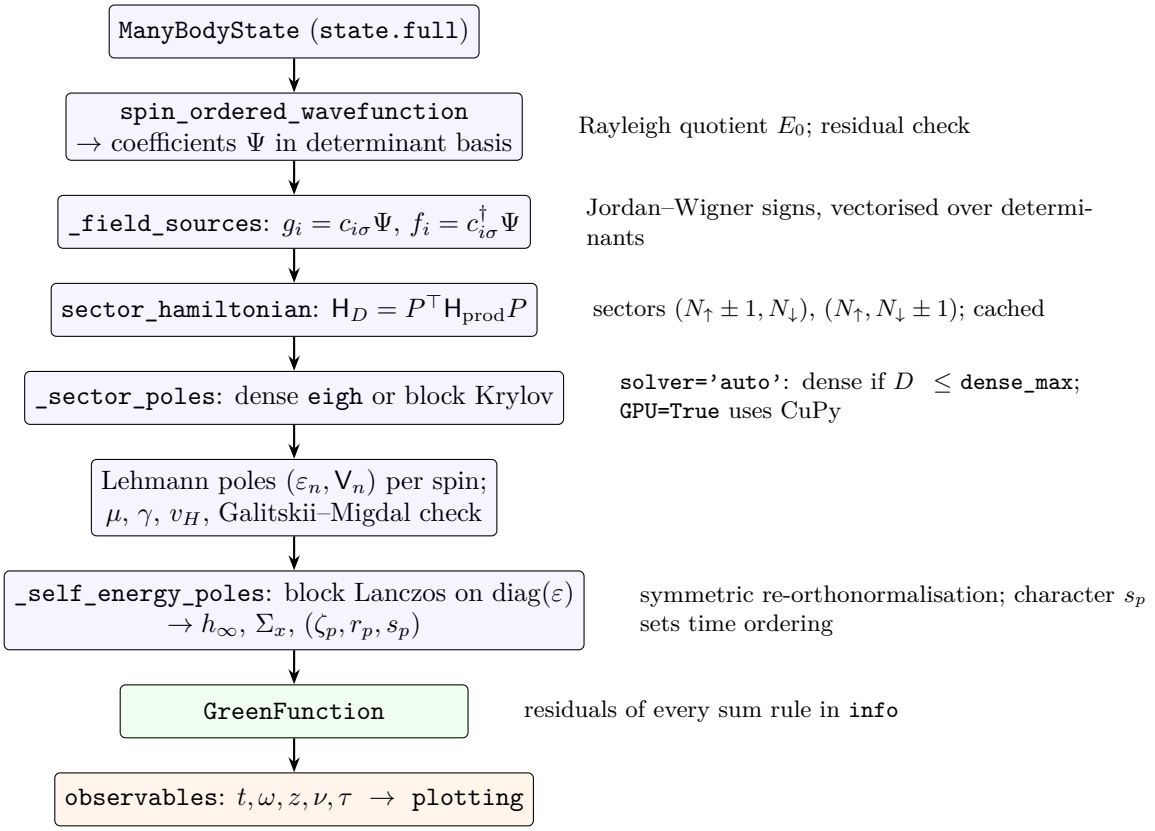


Figure 2: Data flow of `iDEA.methods.interacting.green_function` and its consumers.

- $|\operatorname{Tr} G(t)|$ for real-time G;
- $\operatorname{Re} \operatorname{Tr} \Sigma$ for the self-energy.

Videos are written as `.mp4` (FFmpeg) or `.gif` (Pillow).

Visual aids for real time. A point source $c_i^\dagger |\Psi\rangle$ excites every energy on the grid, up to $\sim 2/\delta x^2$, so real-time quantities are dominated by fast grid modes. Two optional aids make real-time videos readable:

- `cutoff` weights each pole by $e^{-(\varepsilon_n-\mu)^2/2c^2}$, a smooth energy filter. The result is then no longer exact, and the documentation says so.
- `demodulate` multiplies by $e^{i\mu t}$ to remove the trivial overall phase rotation.

4 Implementation

4.1 Module map and data flow

The new code follows iDEA's existing split between methods, states, observables and (now) plotting. Table 3 lists the public functions and Figure 2 the data flow. Two design rules hold throughout:

- `interacting.py` does not import `observables` (that would be a circular import); it computes γ, n and v_H itself.
- The only part of a many-body state that is used is its full wavefunction.

Table 3: Public functions and classes added.

function	purpose
`iDEA/methods/interacting.py`	
`spin_ordered_wavefunction(s, state)`	spatial wavefunction with up electrons first, Equation (19)
`sector_isometry(Nx, n_up, n_down)`	sparse isometry P from determinants to the product space
`sector_hamiltonian(s, n_up, n_down)`	$\mathsf{H}_D = P^\top \mathsf{H}_{\text{prod}} P$
`green_function(s, state, solver, dense_max, krylov_dim, tol, self_energy, GPU, silent)`	exact G and Σ_{xc} in pole form
`iDEA/state.py`	
`GreenFunction`	container: `mu`, `energy`, `v_h`, `info`; per spin `energies`, `orbitals`, `occupations`, `sigma_x`, `sigma_energies`, `sigma_orbitals`, `sigma_character`
`iDEA/observables.py`	
`green_function(s, state, t|omega|z|nu|tau, eta, kind, spin, cutoff)`	G on any axis; accepts `GreenFunction`, `ManyBodyState`, `SingleBodyState`, `QuasiParticleState`
`self_energy(s, state, ..., part, method, spin, cutoff)`	Σ_{xc}, Σ_c or Σ_x from poles or by Dyson inversion
`spectral_function(s, state, omega, eta, spin, diagonal)`	$A(x, x'; \omega)$ or $A(x, x; \omega)$
`galitskii_migdal_energy(s, state)`	total energy from G, Equation (10)
`iDEA/plotting.py` (new)	
`plot_two_point`, `animate`	snapshots and videos of any $A(x, x')$
`animate_green_function`, `animate_self_energy`	videos of G and Σ with traces

4.2 The algorithm

Listing 1 summarises `green_function`, and Listing 2 the shared block Krylov routine, which serves both the sector solve and the self-energy partition.

4.3 Diagnostics

`GreenFunction.info` records:

- `rayleigh_residual`: the Rayleigh residual of the state;
- `sectors`: for each sector, the solver used, its dimension and its Krylov dimension;
- `{up,down}_completeness_error`: $\max |\mathsf{V}\mathsf{V}^\dagger - 1|$;
- `{up,down}_first_moment_error`: $\max |\Sigma_x - (-\gamma_\sigma v)|$;
- `galitskii_migdal_error`: $|E_{\text{GM}} - E_0|$.

Every one of these should be at machine precision. A user can therefore check any calculation, including truncated Krylov runs, from `info` alone.

4.4 Cost

- **Building a sector.** This costs one sparse product $P^\top \mathsf{H}_{\text{prod}} P$, with $n_\uparrow!\, n_\downarrow!$ non-zeros per column of P.

```
ψ = spin_ordered_wavefunction(s, state)                    # checks Σ|ψ|² = 1
Ψ = √(N↑!N↓!) ψ[determinants];  E0 = ⟨Ψ|H_D|Ψ⟩;  check residual
for each spin σ:
    for removal (c_iσ) and addition (c†_iσ):
        H = sector_hamiltonian(sector)                     # cached per sector
        B = _field_sources(Ψ, σ, create)                   # D × N_x
        (E_n, ⟨n|B⟩) = dense eigh  or  block Krylov        # _sector_poles
        keep poles with weight > 10^-14;  ε_n = ±(E_n − E0);  V = overlaps
μ = midgap over both spins (or None);  γ_σ, n, v_H from removal poles
for each spin σ:
    (h_∞, ζ, r, s) = _self_energy_poles(ε, V, character)
    Σ_x = (h_∞ − h0 − diag(v_H))/δx;   first-moment error = max|Σ_x − (−γ_σ v)|
    warn about poles with |s_p| < 10^-6
Galitskii-Migdal error = |E_GM − E0|;  return GreenFunction
```

Listing 1: `interacting.green_function` (simplified).

```
Q1 = left singular vectors of B with σ > tol·σ_max        # or B itself if orthonormal
repeat:
    W = apply(Q_last)                                     # H_D Q (sparse) or E Q (diagonal)
    if dimension reached: break
    W ← W − Q(Q†W)  twice                                 # CGS2
    Q_new = left singular vectors of W with σ > tol·scale  # deflation
    if Q_new is empty: break                              # space exhausted: exact
T = Q†(HQ), symmetrised;   return T, Q
```

Listing 2: `interacting._block_krylov(apply, B, max_dim, tol, scale)`.

- **Dense diagonalisation.** $O(D^3)$ time and $O(D^2)$ memory: $D = 22\,801$ needs 4.2 GB per double-precision copy.
- **Block Krylov with K vectors.** $O(K\,\mathrm{nnz}(\mathsf{H}_D) + DK^2)$ time and $2DK$ numbers of memory.
- **The self-energy stage.** A spin channel with M poles needs $O(M^3)$ time and $O(M^2)$ memory, on the CPU. This becomes the most memory-hungry step for the largest dense runs, with about 20 GB for $M \approx 23\,000$.
- **Evaluation.** $O(N_x^2 M)$ per time or frequency point.

Measured times. On the container used for development (CPU only):

- hydrogen at $N_x = 61$ (sectors of 1,830 and 3,721, dense) takes 19 s;
- a two-electron atom at $N_x = 41$ (sectors of 33 620, Krylov dimension 1 230) takes 17 s;
- the full test suite takes about 50 s, and about 10 s on the GitHub runner.

4.5 Usage

Listing 3 shows a complete calculation.

The hydrogen example runs end to end in the command-line script `tutorial/green_function_hydrogen.py` and in the notebook `tutorial/example_green_function.ipynb`. The script `report/make_figures.py` regenerates every figure and number in this report.

5 Testing and validation

Before writing the code. Every formula of Section 3 was first checked against an independent brute-force implementation in the full Fock space of small lattices. That implementation used

```
import numpy as np, iDEA

x = np.linspace(-12, 12, 61)
v_int = iDEA.interactions.softened_interaction(x)
s = iDEA.system.System(x, -1.0 / (abs(x) + 1.0), v_int, "u")
state = iDEA.methods.interacting.solve(s, k=0)
green = iDEA.methods.interacting.green_function(s, state)  # solver="auto", GPU=False
print(green.info)                                           # sum-rule residuals

omega = np.linspace(-1.0, 2.0, 150)
G = iDEA.observables.green_function(s, green, omega=omega, eta=0.05)      # G(x,x';w)
S = iDEA.observables.self_energy(s, green, omega=omega)                   # Sigma_xc(x,x';w)
Sc_t = iDEA.observables.self_energy(s, green, t=np.linspace(0, 20, 200), part="c")
S_iw = iDEA.observables.self_energy(s, green, nu=np.linspace(0, 3, 50))   # z = mu + i nu
E = iDEA.observables.galitskii_migdal_energy(s, green)                    # equals state.energy

gw = iDEA.methods.mbpt.solve(s, mode="GW")                  # the same calls work for GW
S_gw = iDEA.observables.self_energy(s, gw, nu=np.linspace(0, 3, 50))
iDEA.plotting.animate_self_energy(s, green, omega=omega, spin="down",
                                  file_name="Sigma.mp4")
```

Listing 3: Computing, evaluating and animating the exact G and Σ of the hydrogen atom, and the same calls for GW.

explicit matrices for c_i and $c_i^\dagger$ and covered:

- completeness;
- the first moment;
- Löwdin partitioning against direct Dyson inversion;
- the Galitskii–Migdal energy for two and three electrons;
- the Jordan–Wigner formulas.

All agreed to better than 4×10^{-15}. An independent review re-derived the whole pipeline in NumPy on small `u`, `ud`, `uu` and `du` grids, and found every sum rule to be at most 3×10^{-15} and the hydrogen self-interaction identities at about 10^{-16}.

The test suite. The suite `tests/test_green_function.py` (Table 4) uses grids of 9–15 points, so that every sector can also be diagonalised completely. It covers:

- one electron (hydrogen, `u`);
- two electrons (a helium-like singlet, `ud`);
- an excited triplet with no gap (`uu`);
- three electrons (lithium-like, `uud`).

It runs in the continuous-integration workflow on every push. The 30 CPU tests pass, and the 2 GPU tests are skipped without CuPy.

The production example. The hydrogen calculation of Section 6 ($N_x = 61$) reports:

- a Rayleigh residual of 7×10^{-15};
- a completeness error of 1×10^{-15};
- a first-moment error of 6×10^{-14};
- a Galitskii–Migdal error of 3×10^{-16};

Table 4: The tests in `tests/test_green_function.py`.

test	identity checked	tolerance
`test_sum_rules` (u, ud, uu, uud)	Rayleigh residual, completeness, first moment (Σ_x is Fock of exact γ), Galitskii–Migdal, E_0	10^{-10}–10^{-8}
`test_field_operators`	$\langle g_i \vert g_j\rangle = \gamma_{ji}\delta x$, $\langle f_i \vert f_j\rangle = \delta_{ij} - \gamma_{ij}\delta x$ (Equation (26))	10^{-12}
`test_density_matrix`	$G(t{=}0^-) = i\gamma$ and $G(\tau{=}0^-) = \gamma$ against `observables.density_matrix`	10^{-12}
`test_time_jump`	$G(0^+) - G(0^-) = -i\delta_{ij}/\delta x$	10^{-8}
`test_galitskii_migdal`	`galitskii_migdal_energy` $= E_0$ (u, ud, uud)	10^{-10}
`test_non_interacting_limit`	$v = 0$: $\Sigma_{xc} = 0$ and $G = G_0$ of `non_interacting.solve`	10^{-9}
`test_consistency_with_mbpt`	$v = 0$: $G(i\nu)$, $G(\tau)$ equal `mbpt.non_interacting_green_function` on its grids	10^{-9}
`test_exhausted_krylov_is_exact`	Krylov with an exhausted space = dense, for G and Σ	10^{-9}
`test_truncated_krylov_moments`	three blocks: moments $0 \ldots 5$ exact, sum rules exact	10^{-8} rel.
`test_poles_match_dyson` (ud, uu, uud)	pole-form Σ_{xc} = Dyson inversion at complex z and retarded real ω	10^{-9}
`test_matches_loewdin_partitioning`	Σ_c = Schur complement with a QR completion of V	10^{-9}
`test_exchange_is_fock`	Σ_x = `observables.exchange_potential` of exact γ	10^{-9}
`test_time_ordering` (ud, uu)	$\operatorname{sgn} \operatorname{Im} \langle r_p \vert \Sigma^T_{\mathrm{Dyson}}(\zeta_p) \vert r_p\rangle = -\operatorname{sgn} s_p$ for isolated poles	sign
`test_hydrogen_self_interaction`	single removal pole at E_0; $(v_H + \Sigma_x)\phi = 0$; $\Sigma_c(z)\phi = 0$; all $s_p = +1$	10^{-9}
`test_imaginary_time_matches_fourier_transform`	$G(\tau)$ = `SpaceTimeGrid` transform of $G(i\nu)$	10^{-6}
`test_quasiparticle_state`	a GW state passes through `green_function`, `self_energy`	10^{-6}
`test_errors`	invalid axes and parts; no gap $\Rightarrow$ μ is `None` and `nu` is refused	raises
`test_invalid_solver`	unknown solver	raises
`test_normalisation`	$\int A(x, x; \omega)\, d\omega\, \delta x = 1$	5×10^{-3}
`test_matches_retarded_green_function`	$A = \frac{i}{2\pi}(G^R - G^{R\dagger})$	10^{-12}
`test_animations`	`.gif` written for $G(\omega)$ and $\Sigma_c(t)$; snapshot; invalid input	runs
`test_gpu_matches_cpu` (dense, Krylov)	GPU = CPU for G and Σ (skipped without CuPy)	10^{-9}

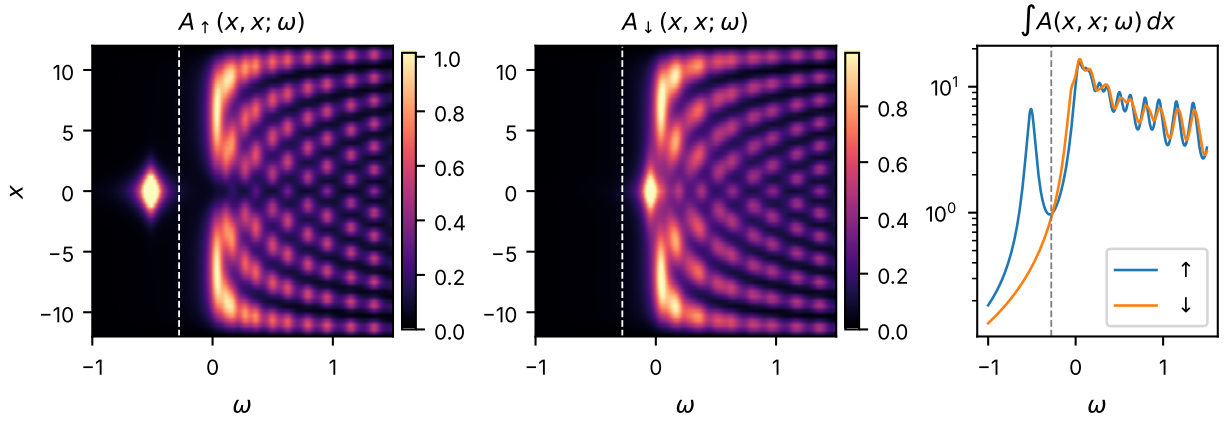


Figure 3: Local spectral functions $A_\sigma(x, x; \omega)$ of the hydrogen atom ($\eta = 0.05$) and their traces. The dashed line marks μ. Spin up: the removal pole at E_0 and the triplet continuum. Spin down: the singlet continuum, with the H^- bound state at its threshold.

Table 5: Hydrogen at $N_x = 61$: total (Galitskii–Migdal) energies and removal energies. For GW, the removal energy is the static quasiparticle energy of the occupied state.

method	total energy	removal energy
exact	−0.510834	−0.510834
G_0W_0@HF	−0.517064	−0.5165
GW_0	−0.516250	−0.5150
GW	−0.516269	−0.5138

- $\|(v_H + \Sigma_x)\phi\|_\infty = 1 \times 10^{-14}$.

6 Illustrative results: the one-dimensional hydrogen atom

The system. One spin-up electron in $v_{\text{ext}} = -1/(|x| + 1)$, with the softened interaction $v(x, x') = 1/(|x - x'| + 1)$, on $N_x = 61$ points in $[-12, 12]$ ($\delta x = 0.40$). The exact ground-state energy is $E_0 = -0.510834$. The midgap chemical potential is $\mu = -0.2784$. The spin-up and spin-down channels have 1,831 and 3,721 poles. Every number in this section is regenerated by `report/make_figures.py` (Appendix A).

Spectral functions (Figure 3). In the spin-up channel, the single removal pole at E_0 carries the whole occupied orbital. In the spin-down channel, an electron added to the atom can bind as H^-: there is an addition pole at -0.0460, just below the continuum, so the electron affinity is 0.0460. Above it, the box states of the continuum appear as the regular pattern of standing waves.

Self-interaction and self-screening (Figure 4, Table 5). The exact $\Sigma_{c\uparrow}(\mu + i\nu)$ annihilates the occupied orbital to 2×10^{-17} at every frequency (Section 2.4). G_0W_0@HF, GW_0 and self-consistent GW leave residuals of 1.7×10^{-2}, 1.7×10^{-2} and 1.7×10^{-2}: the self-screening error [8, 11]. In energy terms, every GW variant overbinds the electron, both in the total energy and in the ionisation energy.

The static correlation self-energy (Figure 5). The full two-point structure of $\Sigma_c(x, x'; \mu+i0)$ is shown in Figure 5.

- **Spin up.** The exact $\Sigma_{c\uparrow}$ is not zero. It acts only in the space orthogonal to ϕ, and correlates a second spin-up electron with the first. Its most negative element is -0.0061, half that of

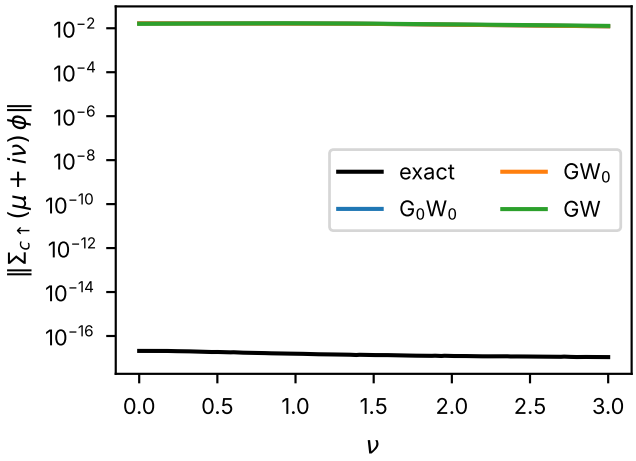


Figure 4: Self-screening: $\|\Sigma_{c\uparrow}(\mu + i\nu)\phi\|$ for the exact self-energy (black; machine zero) and for GW at three levels of self-consistency, each with its own occupied orbital.

GW (-0.0116). The GW $\Sigma_{c\uparrow}$ differs from it near the nucleus, where the occupied orbital lives. This difference is the self-screening of Figure 4.

- **Spin down.** The exact $\Sigma_{c\downarrow}$ is the polarisation potential felt by an added spin-down electron. Its most negative element is -0.0309, against -0.0220 in GW.

Both self-energies are concentrated near the diagonal and within a few units of the nucleus. This is a first glimpse of the nearsightedness that the research proposal sets out to quantify.

Two-point structure on the real axis (Figure 6). The two rows show different things:

- **Top: $G_\downarrow$ at the H^- pole.** At $\omega = -0.0460$, $\mathrm{Im}\, G_\downarrow$ is dominated by $-d\, d^\dagger/\eta$. It is a picture of the H^- Dyson orbital: diffuse, but bound.
- **Bottom: $\Sigma_{xc\downarrow}$ at $\omega = 0.5$, in the continuum.** The imaginary part is non-zero. An electron added at this energy can scatter into configurations in which the bound electron is excited.

These are single frames of the videos produced by `animate_green_function` and `animate_self_energy`.

An electron injected at the nucleus (Figure 7). The column $G(x, 0; t > 0)$ is the amplitude of an electron injected at $x' = 0$, here band-limited with `cutoff=0.5` (Section 3.9). The two spins behave very differently:

- **Spin up.** Pauli exclusion projects ϕ out of the injected electron. This leaves a hole at the nucleus, and the amplitude starts in two lobes. With no bound triplet state, it flies away, and it returns only after reflecting off the box.
- **Spin down.** Part of the amplitude binds as H^- and stays at the atom, while the rest radiates away.

7 Limitations and outlook

- **System size.** The method is exact but its cost grows exponentially with N. With dense sectors, one electron is routine on fine grids, and two electrons on coarse ones. Block Krylov extends two-electron systems to grids of about 100 points ($N + 1 = 3$), and three electrons to coarser grids.

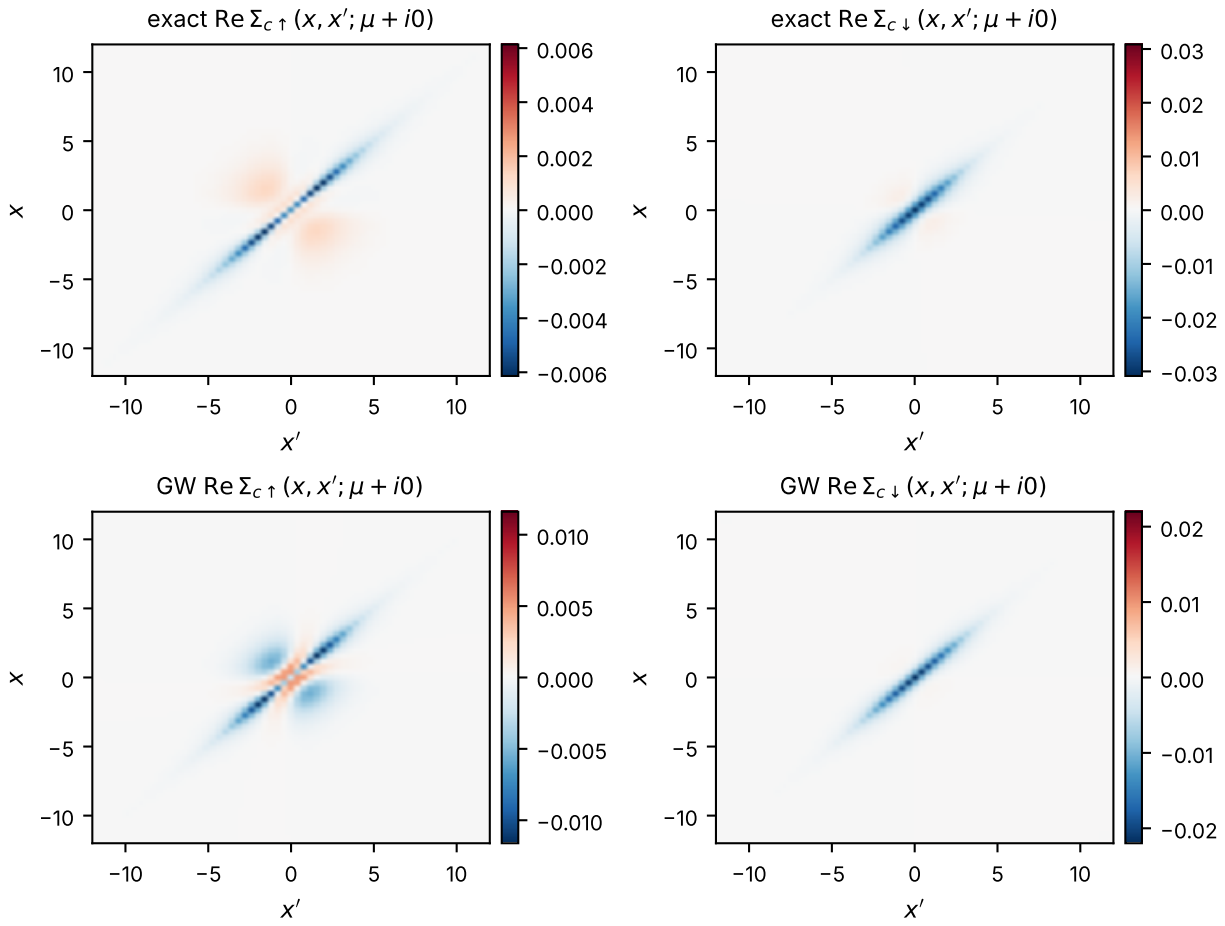


Figure 5: $\mathrm{Re}\,\Sigma_c(x, x'; \mu + i0)$ for both spins, exact (top) and self-consistent GW (bottom). Each panel has its own symmetric colour scale.

- **The self-energy stage.** It runs on the CPU, with $O(M^3)$ time and $O(M^2)$ memory in the number of poles M. A GPU version of this step is a natural extension.
- **Truncated Krylov.** The pole positions change, but the low moments do not (Section 3.5). Properties that depend on fine spectral detail far from μ need convergence checks in `krylov_dim`.
- **Visualisation filter.** `cutoff` is for visualisation only and is not exact.
- **States without a gap.** For states with no gap, such as excited states, μ is `None`, and imaginary-axis evaluation needs an explicit z.
- **Hedin's other functions.** The exact polarisability P and screened interaction W are not constructed. Only G and Σ_{xc} are.

Next steps. These follow the research proposal:

1. production runs on a GPU (Appendix A);
2. two-electron atoms and stretched H_2-like molecules, where Σ develops strong-correlation structure;
3. a quantitative study of the nearsightedness of the exact $\Sigma(x, x'; \omega)$ compared with GW and with exact Kohn–Sham potentials [9].

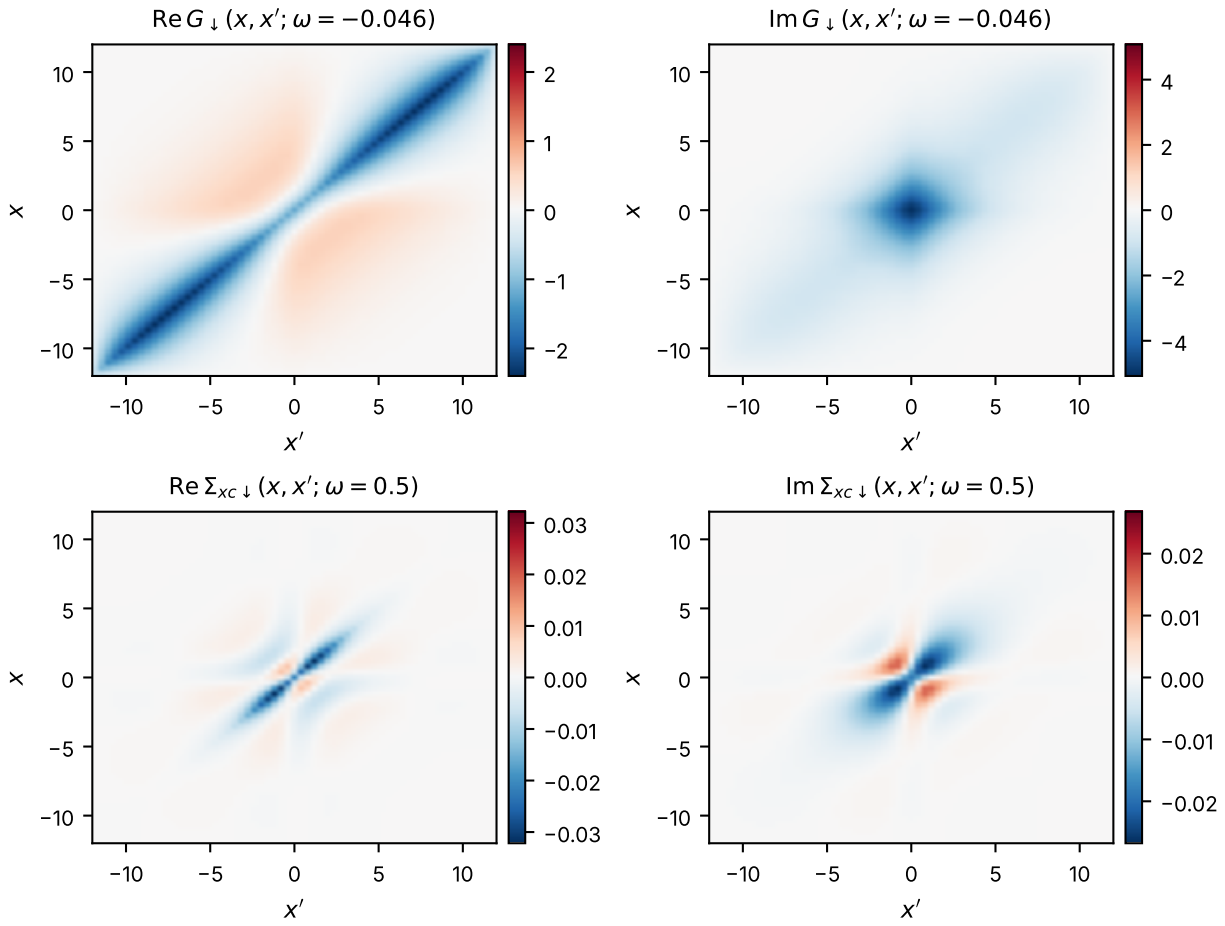


Figure 6: Top: $G_\downarrow(x, x'; \omega)$ at the H^- addition pole. Bottom: $\Sigma_{xc\downarrow}(x, x'; \omega = 0.5)$ in the continuum ($\eta = 0.05$).

A Production calculations on an RTX 4090

These are the runs that were too large for the development container. Install CuPy first with `pip install -e ".[gpu]"`.

1. **GPU tests.** The GPU paths have not yet been run on hardware.

```
pytest tests/test_green_function.py tests/test_gpu.py -v
```

2. **Production hydrogen, with videos.**

```
python tutorial/green_function_hydrogen.py --nx 151 --dense-max 30000 --gpu --out
    hydrogen_151
```

   - The sectors are $(2, 0)$ with $D = 11\,325$ and $(1, 1)$ with $D = 22\,801$. Both are diagonalised densely on the GPU, at about 4.2 GB per copy of the larger matrix; `eigh` needs a few copies.
   - The self-energy stage then runs on the CPU, using about 20 GB of RAM.
   - $N_x = 161$ ($D = 25\,921$) should also fit, but it is tight in 24 GB.

3. **This report at production resolution.**

```
python report/make_figures.py --nx 151 --dense-max 30000 --gpu
```

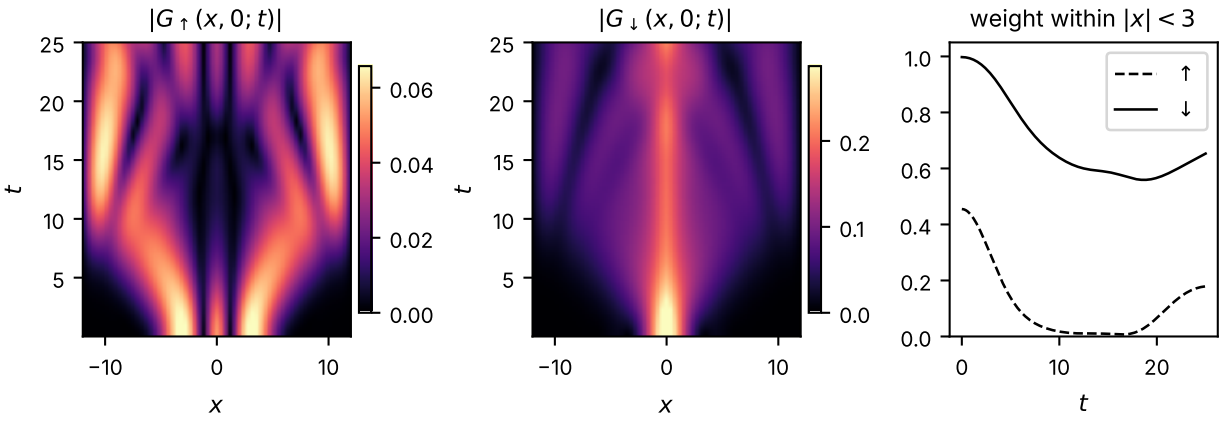


Figure 7: An electron injected at the nucleus: $|G_\sigma(x, 0; t)|$ for spin up and spin down (band-limited with `cutoff=0.5`), and the fraction of $|G|^2$ within $|x| < 3$.

```
make -C report/latex
```

This rewrites `report/latex/figures/*.pdf` and `values.tex`, so every figure and number in Sections 5 and 6 updates.

4. **Convergence.** Repeat at `-nx 101`, and with `-length 20`, to converge the diffuse H^- state. A larger box also delays the reflections in Figure 7.

5. **Optional: two electrons with Krylov.** The $(2, 1)$ and $(1, 2)$ sectors have $D = 510\,050$ at $N_x = 101$:

```
x = np.linspace(-10, 10, 101)
s = iDEA.system.System(x, -2.0 / (abs(x) + 1.0), iDEA.interactions.softened_interaction(x
    ), "ud")
state = iDEA.methods.interacting.solve(s, k=0, GPU=True)
green = iDEA.methods.interacting.green_function(s, state, krylov_dim=1500, GPU=True)
print(green.info)
```

With $K = 1500$, the Krylov basis and its image take about 12 GB on the GPU.

Please send back `info.json` (or `green.info`), the rebuilt `report.pdf` and the best videos.

B Derivations

B.1 Normalisation of the spin-ordered wavefunction

Ψ is antisymmetric under exchange of space–spin pairs and normalised as $\sum_{\{x\sigma\}} |\Psi|^2 \delta x^N = 1$. For each spatial configuration, the spin labels take one of $\binom{N}{N_\uparrow}$ placements of the up spins among the N electrons. Any placement can be brought to the canonical one (ups first) by a permutation of electrons, which changes only the sign. So each placement carries the same weight, $1/\binom{N}{N_\uparrow}$ of the total, and Equation (19) is normalised.

The determinant coefficient $\langle \mathbf{a}; \mathbf{b}|\Psi\rangle$ collects the $n_\uparrow!\, n_\downarrow!$ equal-magnitude orderings of an ordered configuration. With the $1/\sqrt{n_\uparrow! n_\downarrow!}$ normalisation of the antisymmetrised determinant, this gives the factor $\sqrt{n_\uparrow!\, n_\downarrow!}$.

B.2 Jordan–Wigner signs

In $|\mathbf{a}; \mathbf{b}\rangle$, all up operators stand to the left of all down operators, each group in increasing site order. To annihilate $c^\dagger_{a_k\uparrow}$, the operator $c_{a_k\uparrow}$ is anticommuted past the $k - 1$ up operators to its

left, which gives $(-1)^{k-1}$. To annihilate $c^\dagger_{b_k\downarrow}$, it is anticommuted past all $n_\uparrow$ up operators and the $k-1$ down operators before it, which gives $(-1)^{n_\uparrow+k-1}$. The test of Equation (26) checks both the signs and the normalisation.

B.3 The first moment and the exact Fock operator

From the Lehmann representation, $\sum_n \varepsilon_n \mathsf{V}_{in}\mathsf{V}^*_{jn} = \langle\{[c_{i\sigma}, \hat{H}], c^\dagger_{j\sigma}\}\rangle$. For the Hamiltonian of Section 3, $[c_{i\sigma}, \hat{H}] = \sum_k (\mathsf{h}_0)_{ik} c_{k\sigma} + \sum_{k\sigma'} v_{ik}\hat{n}_{k\sigma'} c_{i\sigma}$. The anticommutator with $c^\dagger_{j\sigma}$ gives

$$(\mathsf{h}_0)_{ij} + \delta_{ij}\sum_k v_{ik}\langle\hat{n}_k\rangle - v_{ij}\langle c^\dagger_{j\sigma}c_{i\sigma}\rangle = (\mathsf{h}_0)_{ij} + \delta_{ij}v_H(x_i) - v(x_i,x_j)\gamma_\sigma(x_i,x_j)\,\delta x, \tag{33}$$

which is $h_\infty \delta x$ of Equation (11). The Hartree term contains the total density, and the Fock term the same-spin density matrix. The Fock term exactly removes the self-interaction contained in the Hartree term.

B.4 Galitskii–Migdal in Lehmann form

For a removal pole, $\varepsilon_m = E_0 - E_m$. Since $\hat{H}\,|\Psi_0\rangle = E_0\,|\Psi_0\rangle$, we have $(E_0 - \hat{H})c_{i\sigma}\,|\Psi_0\rangle = [c_{i\sigma}, \hat{H}]\,|\Psi_0\rangle$, so

$$\sum_m \varepsilon_m\, \mathsf{V}^*_{im}\mathsf{V}_{im} = \langle\Psi_0|c^\dagger_{i\sigma}[c_{i\sigma}, \hat{H}]|\Psi_0\rangle\,. \tag{34}$$

Summing over i and σ with the commutator of Appendix B.3 gives $\langle\hat{H}_1\rangle + 2\,\langle\hat{V}\rangle$, where $\hat{H}_1$ is the one-body part and $\hat{V}$ the interaction. Similarly, $\sum_{i\sigma}\sum_m \mathsf{V}^\dagger_m \mathsf{h}_0 \mathsf{V}_m = \langle\hat{H}_1\rangle$. Half the sum of the two is $\langle\hat{H}_1\rangle + \langle\hat{V}\rangle = E_0$, which is Equation (10) [2, 5]. The code evaluates exactly this, $E = \frac{1}{2}\sum_\sigma\sum_m[\mathsf{V}^\dagger_m \mathsf{h}_0 \mathsf{V}_m + \varepsilon_m \|\mathsf{V}_m\|^2]$.

References

[1] L. Hedin. New method for calculating the one-particle Green's function with application to the electron-gas problem. *Phys. Rev.*, 139:A796, 1965.

[2] A. L. Fetter and J. D. Walecka. *Quantum Theory of Many-Particle Systems*. McGraw-Hill, New York, 1971.

[3] G. Stefanucci and R. van Leeuwen. *Nonequilibrium Many-Body Theory of Quantum Systems: A Modern Introduction*. Cambridge University Press, Cambridge, 2013.

[4] R. M. Martin, L. Reining, and D. M. Ceperley. *Interacting Electrons: Theory and Computational Approaches*. Cambridge University Press, Cambridge, 2016.

[5] V. M. Galitskii and A. B. Migdal. Application of quantum field theory methods to the many body problem. *Sov. Phys. JETP*, 7:96, 1958.

[6] M. J. P. Hodgson, J. D. Ramsden, J. B. J. Chapman, P. Lillystone, and R. W. Godby. Exact time-dependent density-functional potentials for strongly correlated tunneling electrons. *Phys. Rev. B*, 88:241102(R), 2013.

[7] M. J. P. Hodgson, J. D. Ramsden, and R. W. Godby. Origin of static and dynamic steps in exact Kohn–Sham potentials. *Phys. Rev. B*, 93:155146, 2016.

[8] J. Wetherell, M. J. P. Hodgson, and R. W. Godby. GW self-screening error and its correction using a local density functional. *Phys. Rev. B*, 97:121102(R), 2018.

[9] J. Wetherell, M. J. P. Hodgson, L. Talirz, and R. W. Godby. Advantageous nearsightedness of many-body perturbation theory contrasted with Kohn–Sham density functional theory. *Phys. Rev. B*, 99:045129, 2019.

[10] P. Romaniello, S. Guyot, and L. Reining. The self-energy beyond GW: local and nonlocal vertex corrections. *J. Chem. Phys.*, 131:154111, 2009.

[11] W. Nelson, P. Bokes, P. Rinke, and R. W. Godby. Self-interaction in Green's-function theory of the hydrogen atom. *Phys. Rev. A*, 75:032505, 2007.

[12] W. Kohn. Density functional and density matrix method scaling linearly with the number of atoms. *Phys. Rev. Lett.*, 76:3168, 1996.

[13] E. Prodan and W. Kohn. Nearsightedness of electronic matter. *Proc. Natl. Acad. Sci. USA*, 102:11635, 2005.

[14] H. Lehmann. Über Eigenschaften von Ausbreitungsfunktionen und Renormierungskonstanten quantisierter Felder. *Nuovo Cimento*, 11:342, 1954.

[15] R. W. Godby, M. Schlüter, and L. J. Sham. Self-energy operators and exchange-correlation potentials in semiconductors. *Phys. Rev. B*, 37:10159, 1988.

[16] H. N. Rojas, R. W. Godby, and R. J. Needs. Space-time method for ab initio calculations of self-energies and dielectric response functions of solids. *Phys. Rev. Lett.*, 74:1827, 1995.

[17] M. M. Rieger, L. Steinbeck, I. D. White, H. N. Rojas, and R. W. Godby. The GW space-time method for the self-energy of large systems. *Comput. Phys. Commun.*, 117:211, 1999.

[18] P. Jordan and E. Wigner. Über das Paulische äquivalenzverbot. *Z. Phys.*, 47:631, 1928.

[19] G. H. Golub and R. Underwood. The block Lanczos method for computing eigenvalues. In J. R. Rice, editor, *Mathematical Software III*. Academic Press, New York, 1977.

[20] E. Dagotto. Correlated electrons in high-temperature superconductors. *Rev. Mod. Phys.*, 66: 763, 1994.

[21] L. Giraud, J. Langou, M. Rozložník, and J. van den Eshof. Rounding error analysis of the classical Gram–Schmidt orthogonalization process. *Numer. Math.*, 101:87, 2005.

[22] G. H. Golub and G. Meurant. *Matrices, Moments and Quadrature with Applications*. Princeton University Press, Princeton, 2010.

[23] O. J. Backhouse, M. Nusspickel, and G. H. Booth. Wave function perspective and efficient truncation of renormalized second-order perturbation theory. *J. Chem. Theory Comput.*, 16: 1090, 2020.

[24] P.-O. Löwdin. Studies in perturbation theory. IV. Solution of eigenvalue problem by projection operator formalism. *J. Math. Phys.*, 3:969, 1962.